Ethanol Blending in Gasoline - Colombia

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U.S. GRAINS & BIOPRODUCTS COUNCIL

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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.

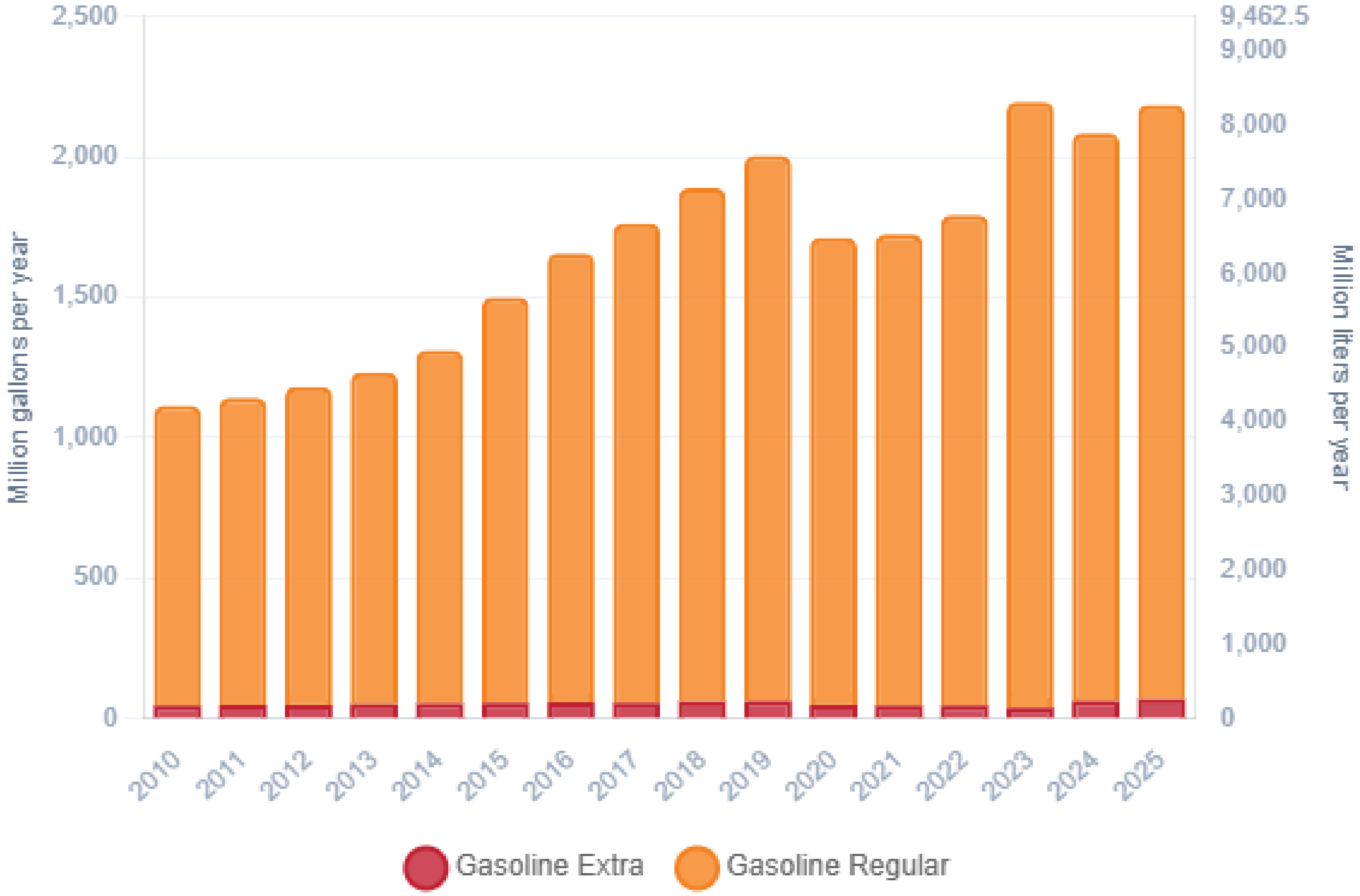
Ethanol Blending in Gasoline - Colombia

In 2025, gasoline consumption in Colombia was 2,184 million gallons (8,270 million liters). Market share was 3% for Extra gasoline E10 (RON 97 – AKI 94) and remaining 97% for Regular gasoline (regular) E10 (RON 89 – AKI 84). Local production supplies 43% of total demand. United States is the main source of gasoline imports.

Ethanol blends are obligatory in cities with more than 500,000 habitants. Resolution 40444 of 2023 authorizes the blending of up to 10% ethanol in the country and allows for lower percentages in particular cases. E10 is the predominant grade in the country except on the border with Venezuela. The ethanol supply comes from imports from the United States and from local production.

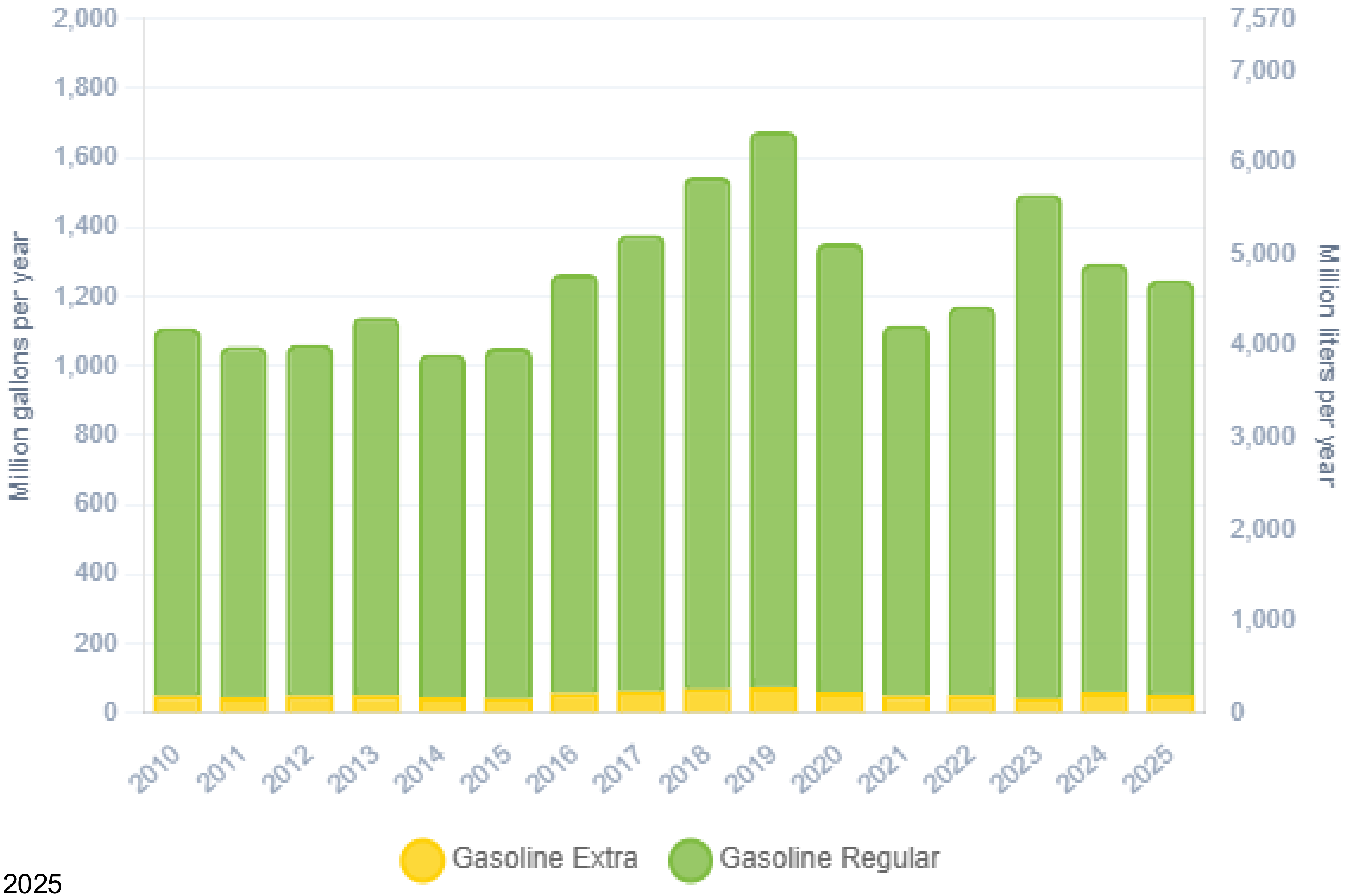
Source: ACP, 2025

Gasoline Demand in Colombia



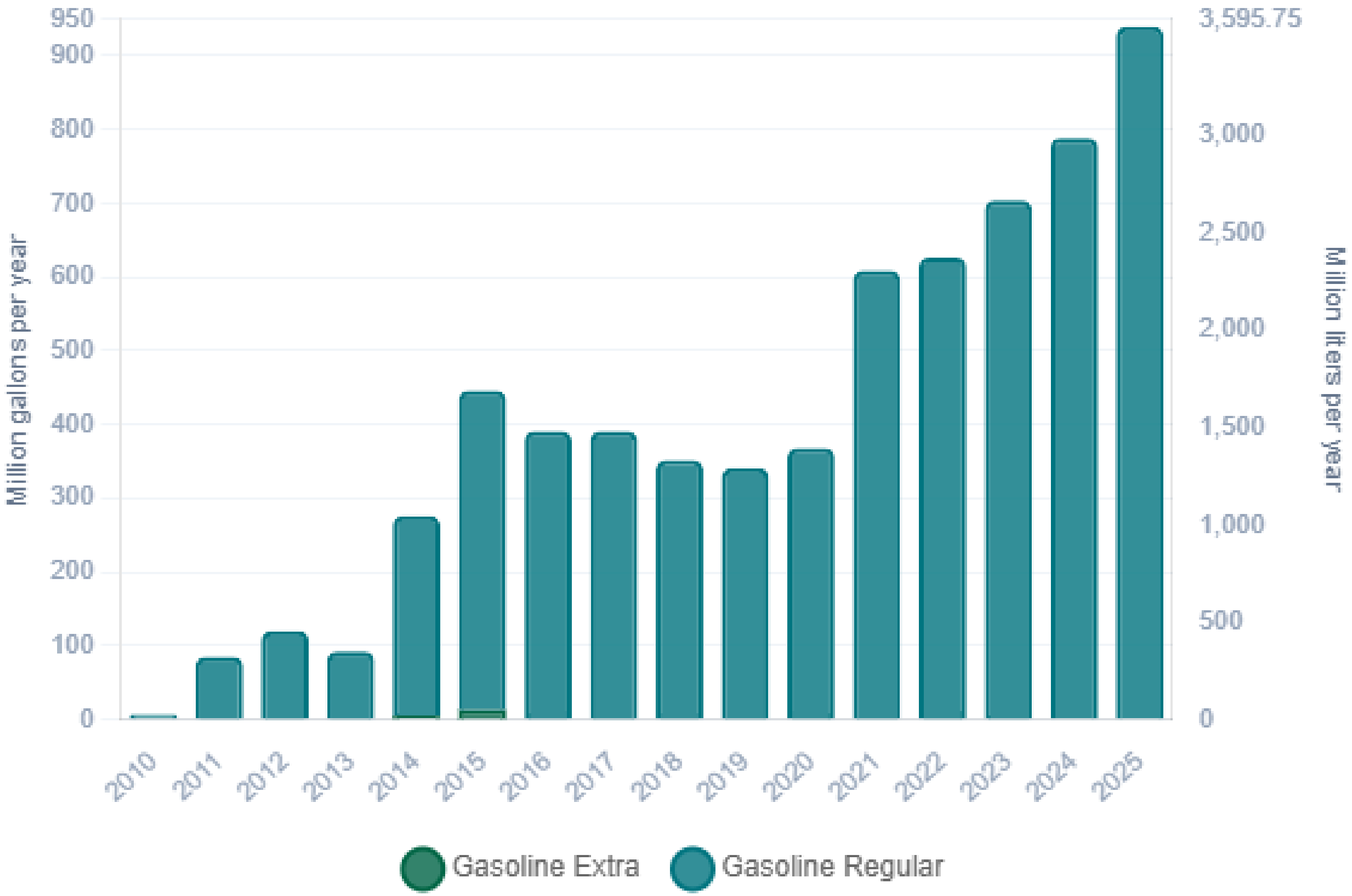
Source: ACP, 2025

Gasoline Production in Colombia



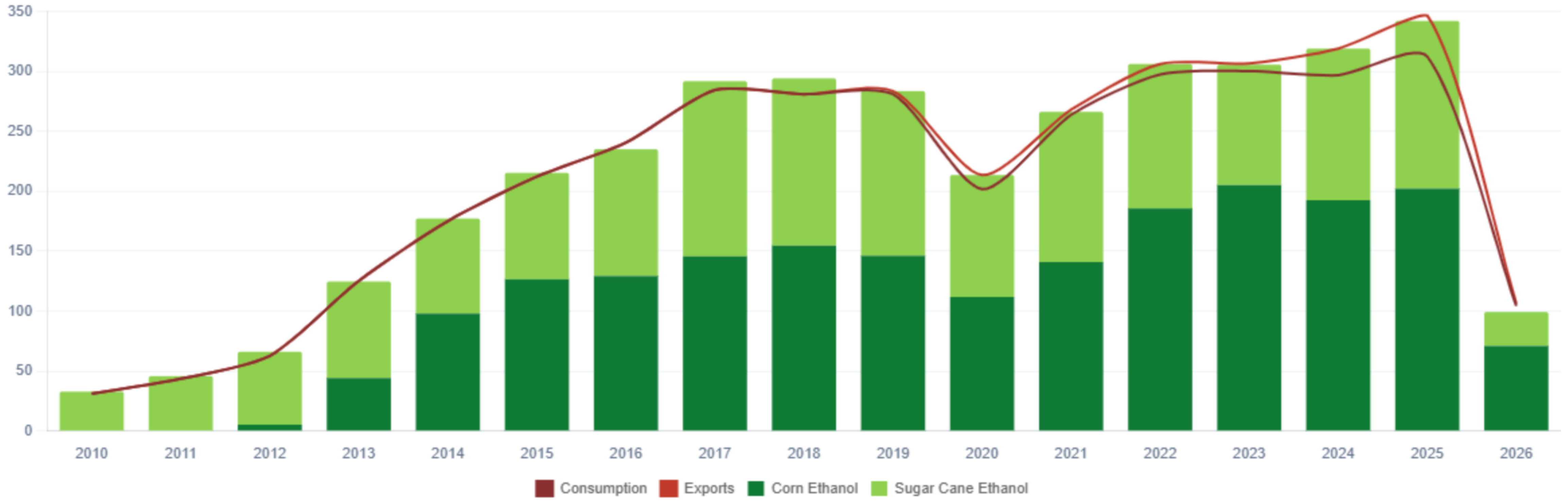
Source: ACP, 2025

Gasoline Imports in Colombia



Source: ACP, 2025

Ethanol Balance in Colombia



Source: Asocaña - Balance Azucarero Colombiano, 2024; EIA, 2024

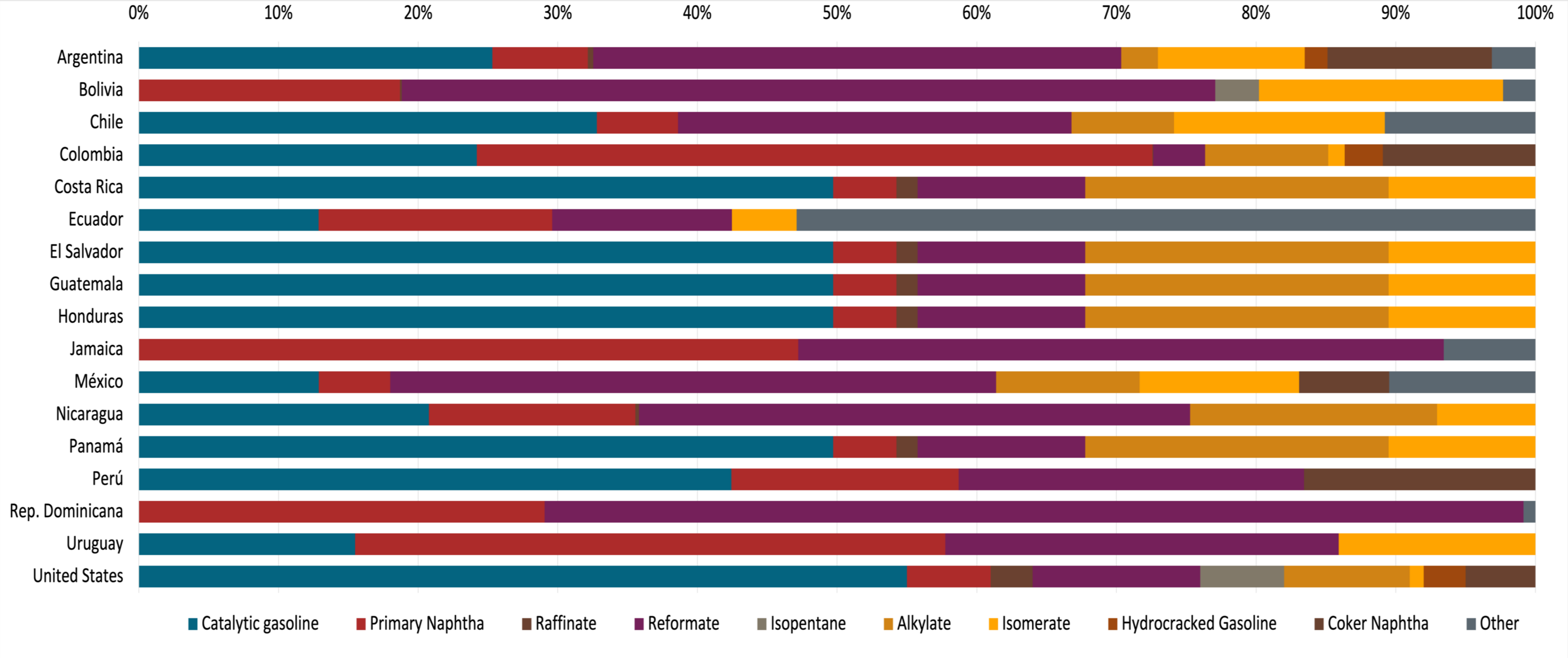
Gasoline Quality in Colombia

Name	Resolution 40103 de 2021				EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	07/01/2023				2017			
Applicability	Whole country	Whole country	Whole country	Whole country	All countries			
Selected Grade	Gasoline Regular	Gasoline Extra	Gasoline Regular E10	Gasoline Extra E10	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1,0 %v/v	< 2,0 %v/v	< 0,9 %v/v	< 1,8 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 28 %v/v	< 35 %v/v	< 25 %v/v	< 31,5 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	-	-	-	-	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	-	-	-	-	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 84	> 93	> 89	> 97	> 95	> 95	> 98	> 98
MON	-	-	-	-	> 85	> 88	> 85	> 88
AKI	> 81	> 91	> 84	> 94				
Sulfur Content	< 50 mg/kg	< 50 mg/kg	< 50 mg/kg	< 50 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content			< 3,7 %m/m	< 3,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	<> 9,5 - 10,5 %v/v	<> 9,5 - 10,5 %v/v	<> 9,5 - 10,5 %v/v	<> 9,5 - 10,5 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<> 65 kPa	<> 65 kPa	<> 65 kPa	<> 65 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)								
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: Ministerio de Energía y Minas, 2021

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



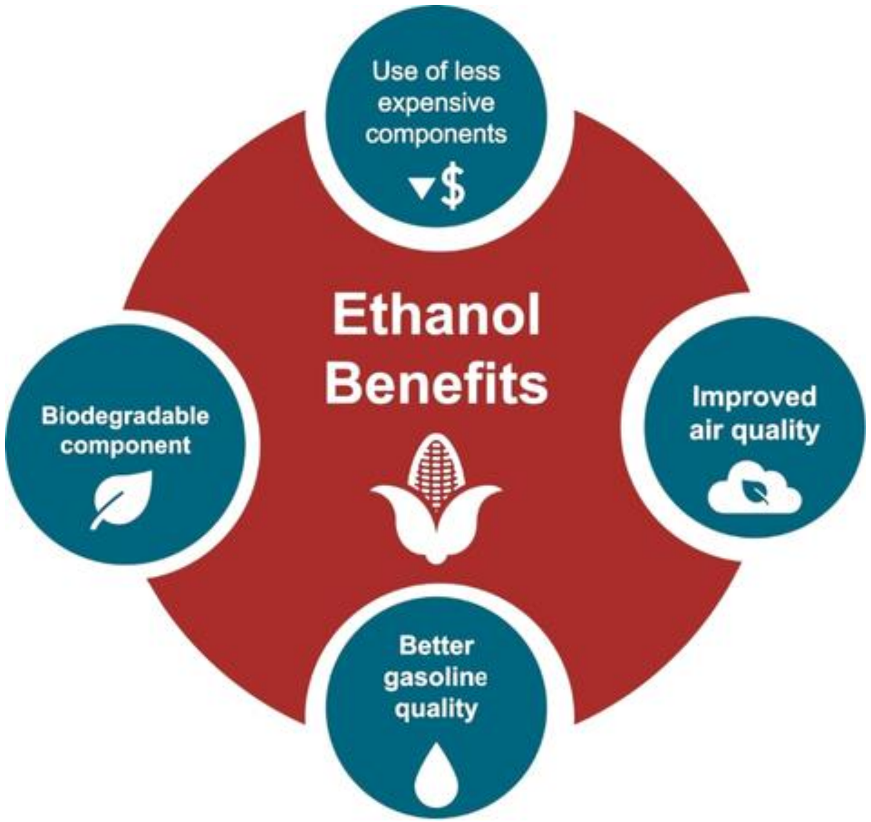
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

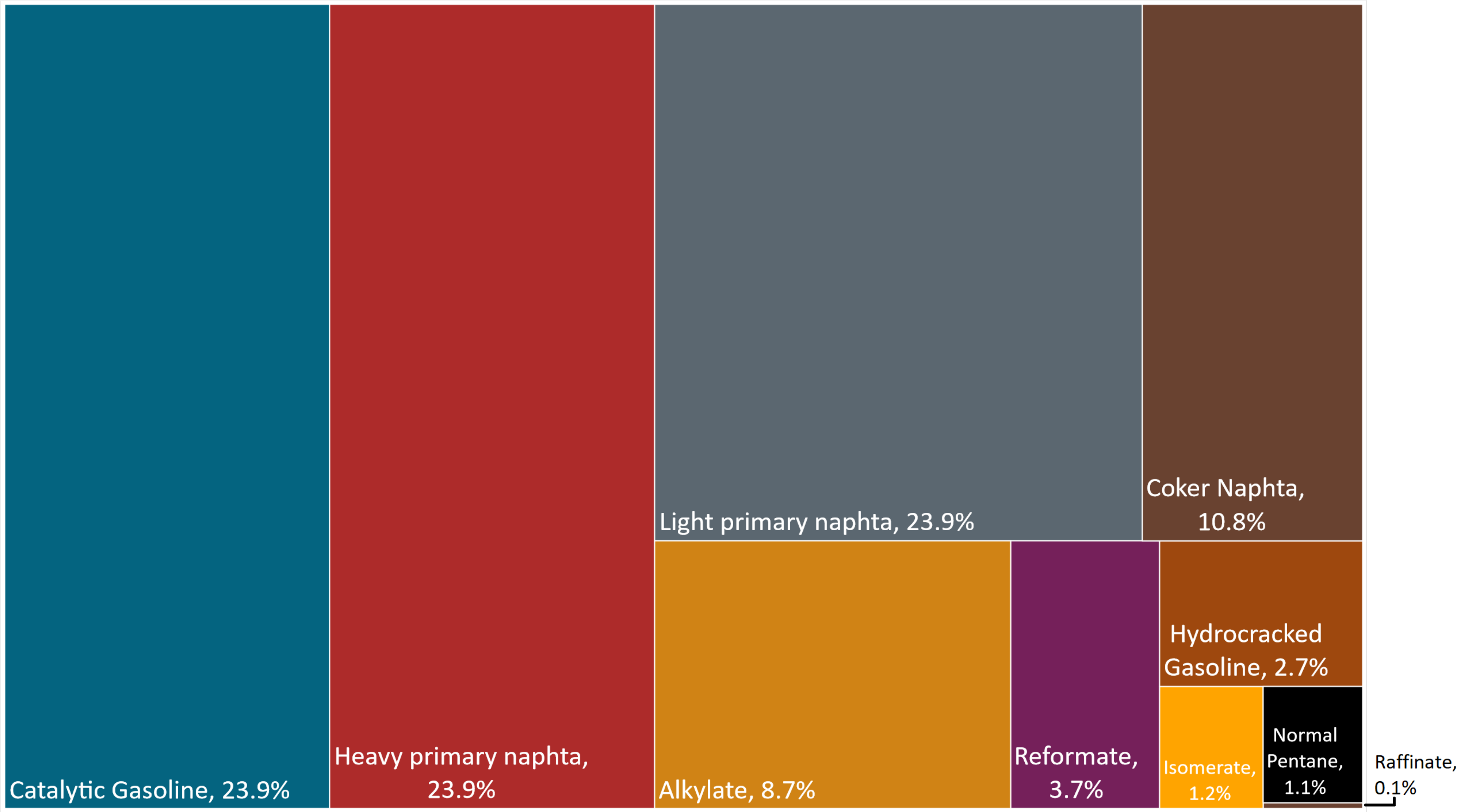
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

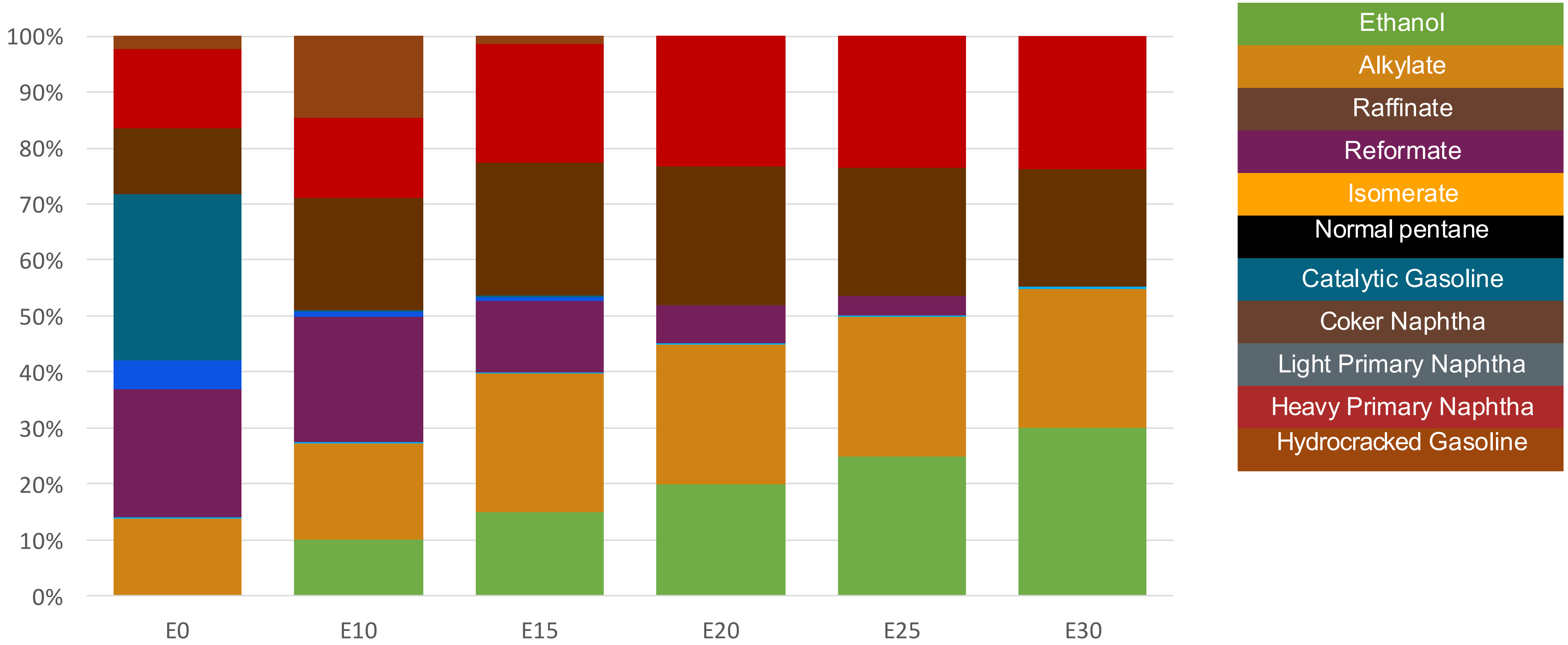


Available Blending Components



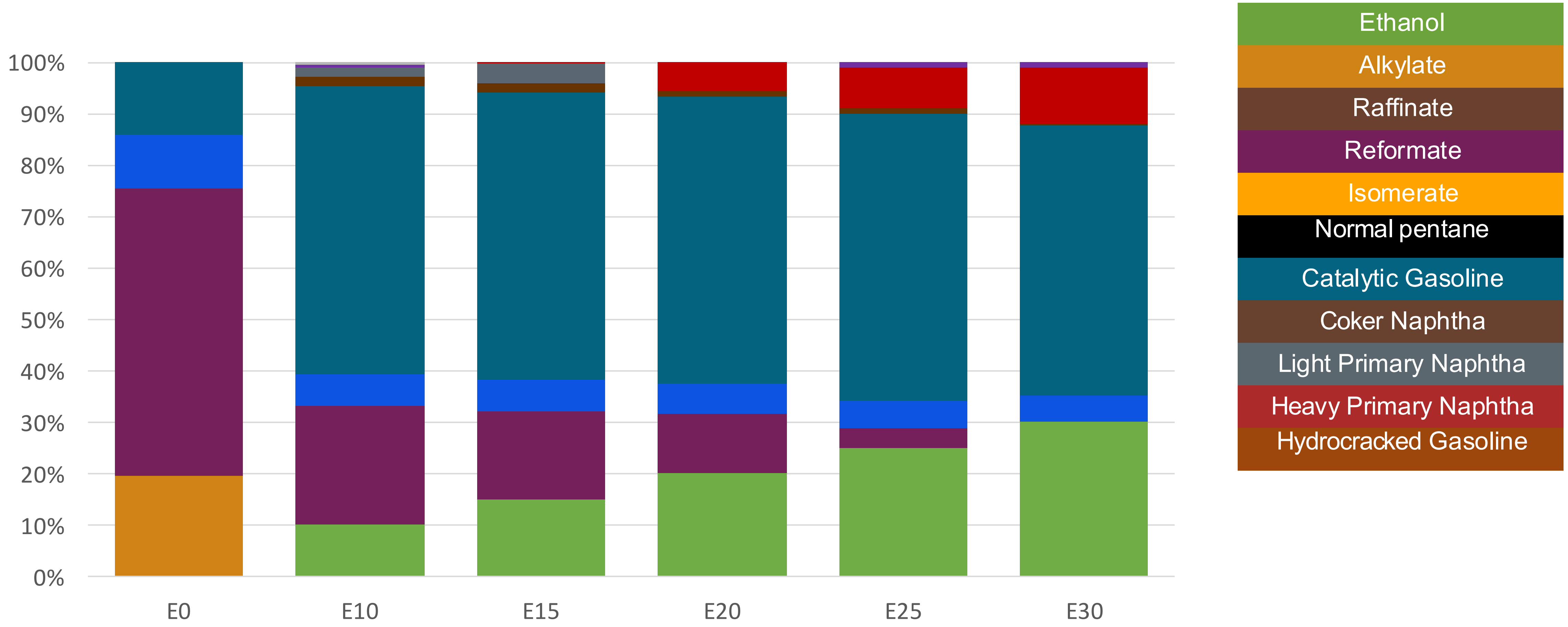
Source: Faro90

Colombia – Regular Gasoline – Constant Octane



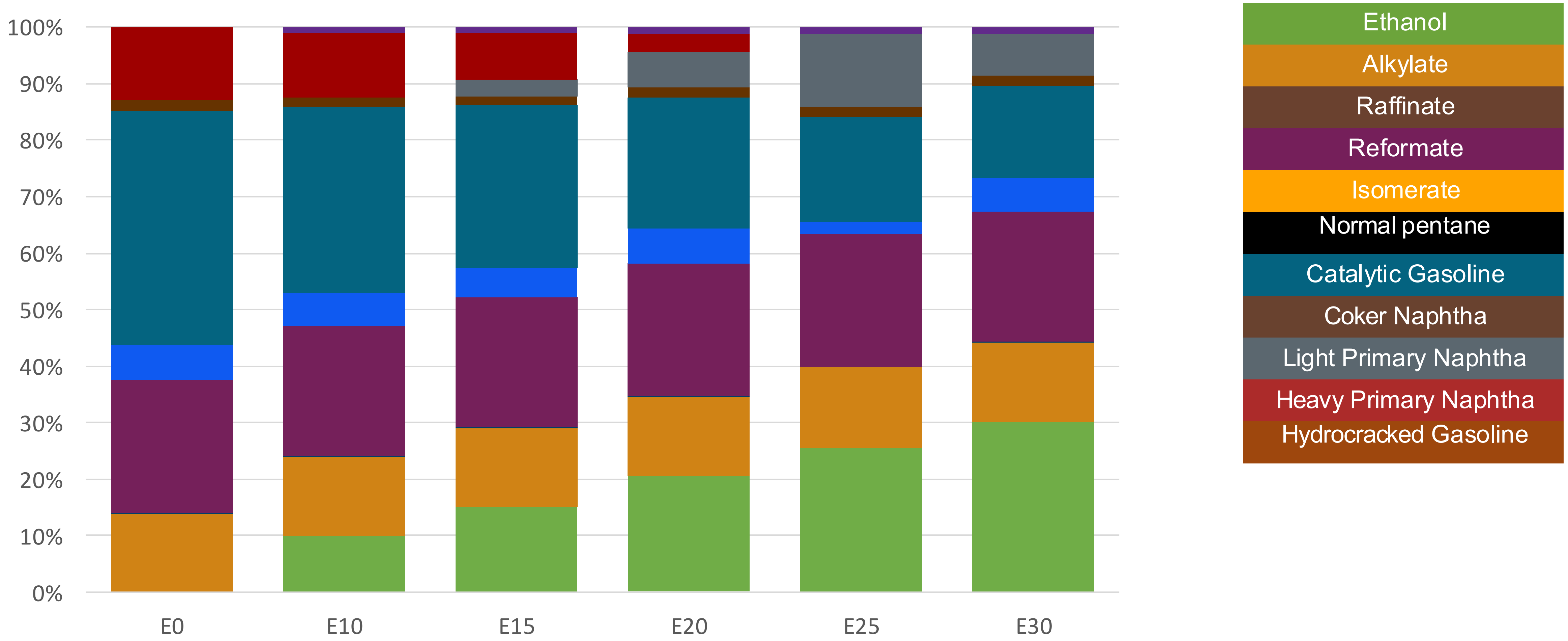
Octane (RON)	89.0	89.0	89.0	89.3	90.3	91.3
Price (US$/gal)	2.049	1.922	1.834	1.760	1.724	1.689

Colombia – Extra Gasoline – Constant Octane



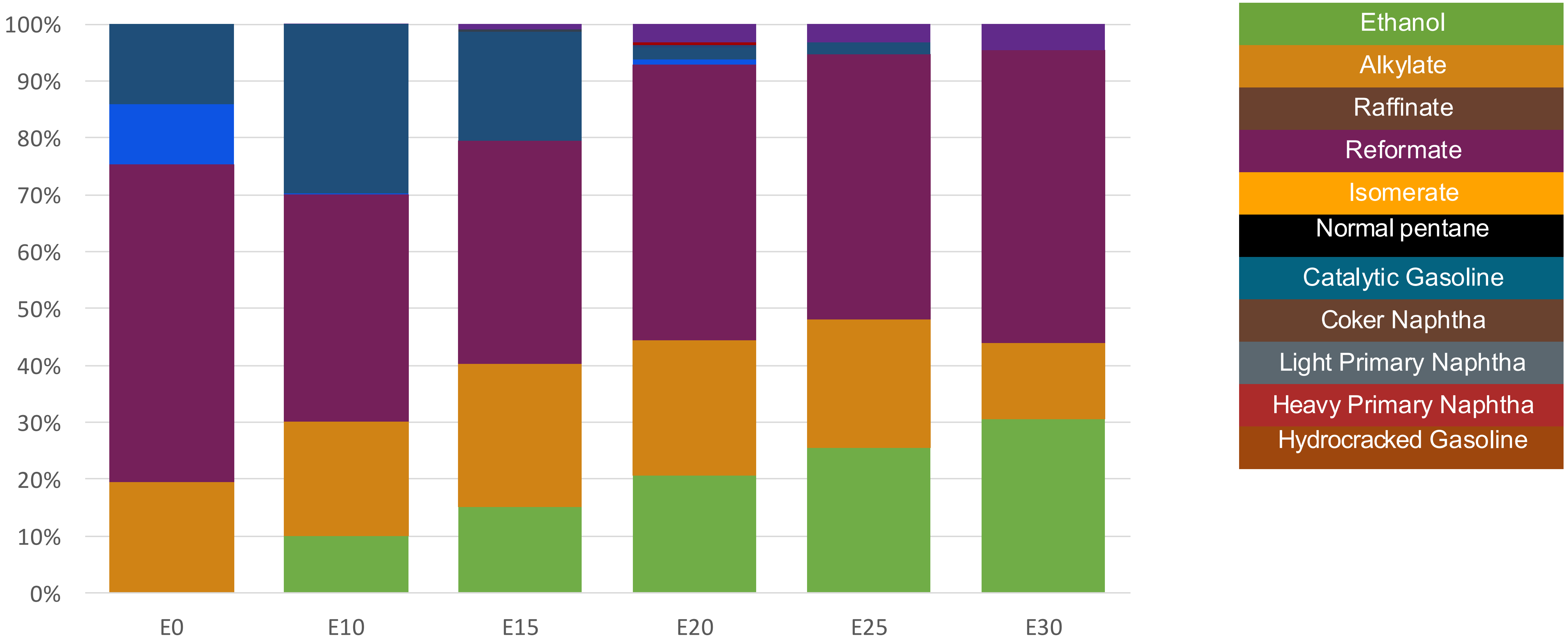
Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US$/gal)	2.449	2.240	2.142	2.053	1.974	1.890

Colombia – Regular Gasoline – Increased Octane



Octane (RON)	90.1	93.5	95.2	97.2	97.9	100.7
Price (US$/gal)	2.117	2.117	2.117	2.117	2.117	2.117

Colombia – Extra Gasoline – Increased Octane



Octane (RON)	97.0	98.2	99.6	101.4	103.0	105.1
Price (US$/gal)	$ 2.449	$ 2.449	$ 2.449	$ 2.449	$ 2.449	$ 2.449

Vehicle Emission Impact for Ethanol Gasoline Blending

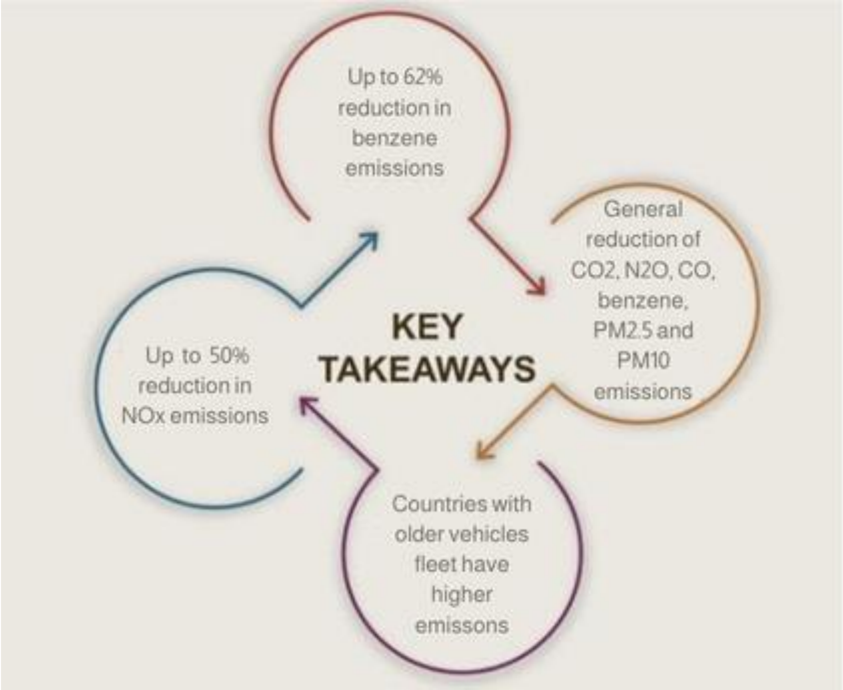
The model used in this analysis takes as a reference the [International Vehicle Emissions Model (IVE).]()
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Colombia

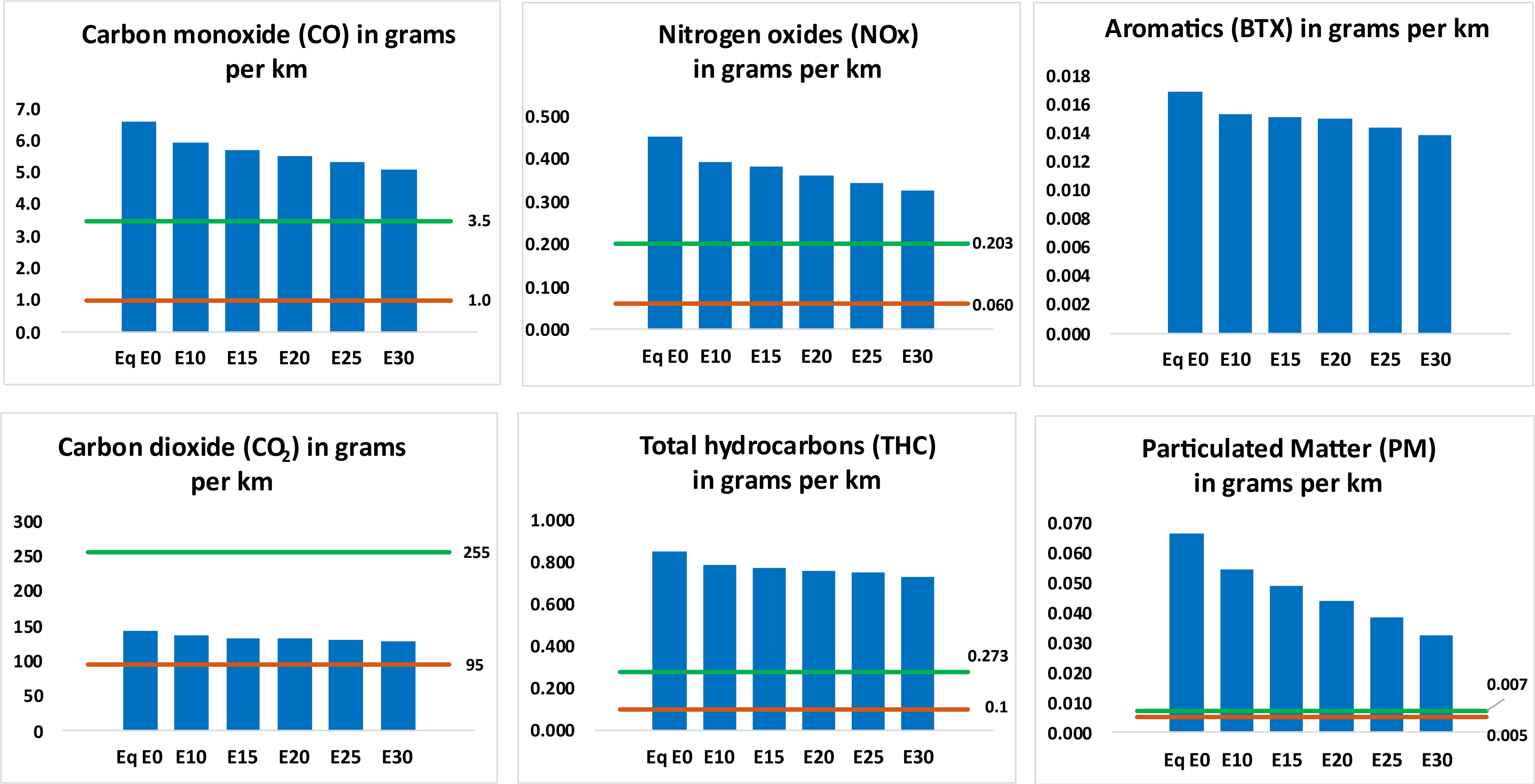
Vehicle Type	Engine	Exhaust	Recovery	Age	Number of Vehicles
Light automobiles	Electric	-	-	Any	19,724
	Multi FI-Pt	Hybrid	PVC Tank	Any	67,899
	Multi FI-Pt	Euro VI		<4 years, < 80 tkm	396440
		Euro V		4-8 yeras, 80-160 tkm	845596
				> 8 years, > 160 tkm	472860
		Euro IV		> 8 years, > 160 tkm	892921
		Euro III		> 8 years, > 160 tkm	1037888
	multi	3-Ways		> 8 years, > 160 tkm	1460739
	single	3-Ways		> 8 years, > 160 tkm	573983
	carbon	3-Ways		> 8 years, > 160 tkm	516661
Medium automobiles	Multi FI-Pt	Euro VI	PCV Tank	<4 years, < 80 tkm	308621
		Euro V		4-8 years, 80-160 tkm	313792
				> 8 years, > 160 tkm	206075
		Euro IV		> 8 years, > 160 tkm	229333
		Euro III		> 8 years, > 160 tkm	206403
	multi	3-Ways		> 8 years, > 160 tkm	61447
	single	3-Ways		> 8 years, > 160 tkm	537
	carbon	3-Ways		> 8 years, > 160 tkm	5
Small Engines	Electric	-	-	Any	3,139
	Multi FI-Pt	Hybrid	PCV	Any	0
	FI 4-ciclos	Euro V		<4 years, < 80 tkm	2507101
				4-8 years, 80-160 tkm	1382788
		Euro IV		> 8 years, > 160 tkm	1816667
				> 8 years, > 160 tkm	626370
		Euro III		> 8 years, > 160 tkm	6119537
		Euro II		> 8 years, > 160 tkm	701662
		none		> 8 years, > 160 tkm	385219
Heavy	Electric	-	-	Any	2,064
	Multi FI-Pt	Hybrid	PCV	Any	1,421
	FI	Euro VI		<4 years, < 80 tkm	55876
		Euro V		4-8 yeras, 80-160 tkm	68512
				> 8 years, > 160 tkm	39235
		Euro IV		> 8 years, > 160 tkm	38784
		Euro III		> 8 years, > 160 tkm	44605
		3-Ways		> 8 years, > 160 tkm	24477

Vehicular Fleet: **17,427,227**
Average Age: **12.8 years**
Motorcycles Fleet: **62%**

Source: RUNT, Andemos, 2025, Faro90

Colombia – Vehicular Emissions

United States
Euro 6



Source: Faro90

Colombia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,805,179	2,668,777	2,532,376	2,436,063	2,346,260	2,279,353	2,182,488	-10%	-16%
CO2	60,851,816	59,287,055	57,809,226	56,646,956	56,071,819	55,804,909	54,776,264	-5%	-8%
VOC	361,584	349,020	336,456	329,133	323,132	318,931	311,867	-7%	-11%
NOx	193,404	178,576	168,261	162,933	154,431	147,729	139,862	-13%	-20%
SOx	695	642	590	545	503	456	408	-15%	-28%
NH3	25,965	25,708	25,450	25,571	25,859	26,061	26,229	-2%	0%
N2O	1,547	1,539	1,533	1,540	1,572	1,589	1,607	-1%	2%
CH4	92,034	92,034	92,034	93,415	95,439	97,188	98,845	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	3,677	3,541	3,404	3,321	3,251	3,200	3,119	-7%	-12%
Acetaldehyde	9,309	10,684	15,676	23,873	32,525	37,166	43,915	68%	249%
Formaldehyde	37,018	35,990	41,954	49,263	51,610	57,095	62,154	13%	39%
Benzene	7,198	6,917	6,555	6,451	6,397	6,118	5,921	-9%	-11%
PM2.5	4,333	3,939	3,545	3,199	2,860	2,496	2,115	-18%	-34%
PM10	24,028	21,843	19,659	17,737	15,856	13,840	11,730	-18%	-34%

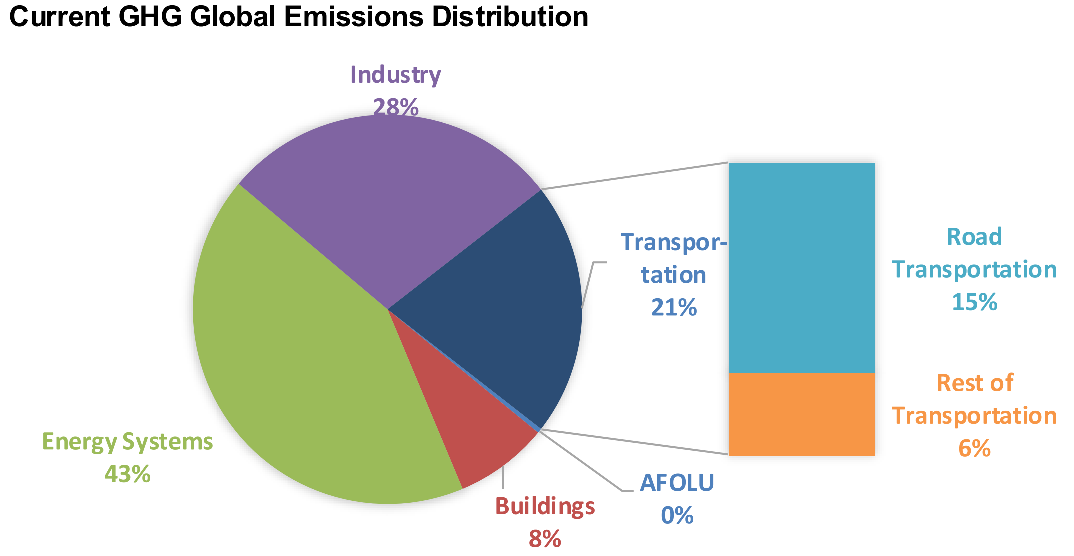
Source: Faro90

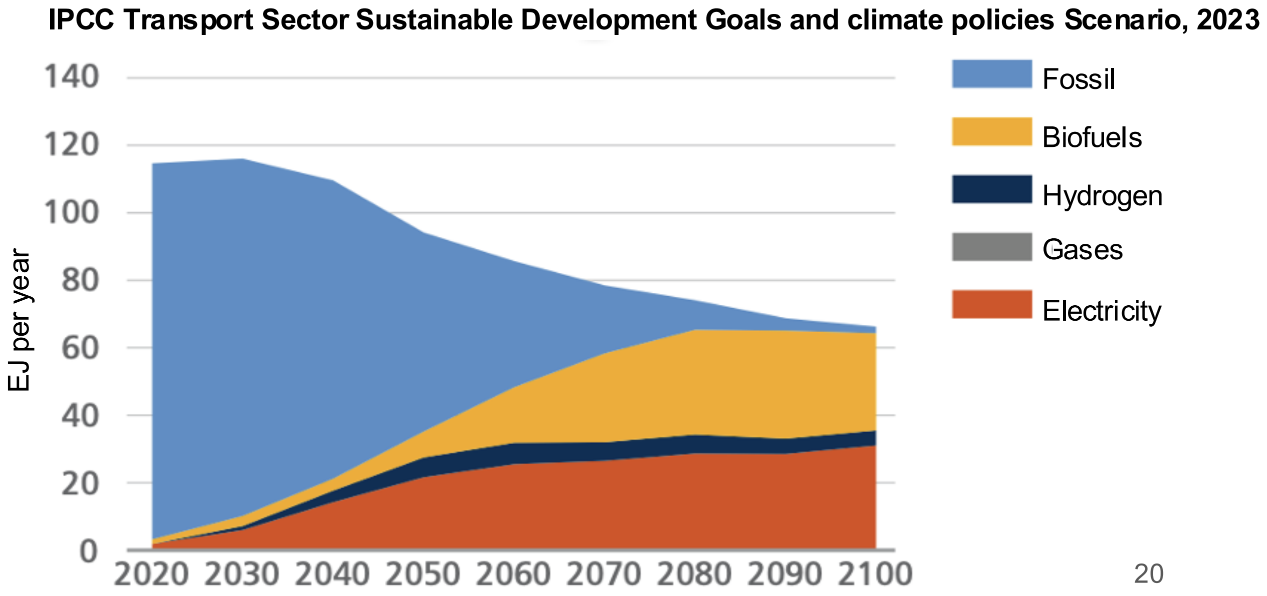
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

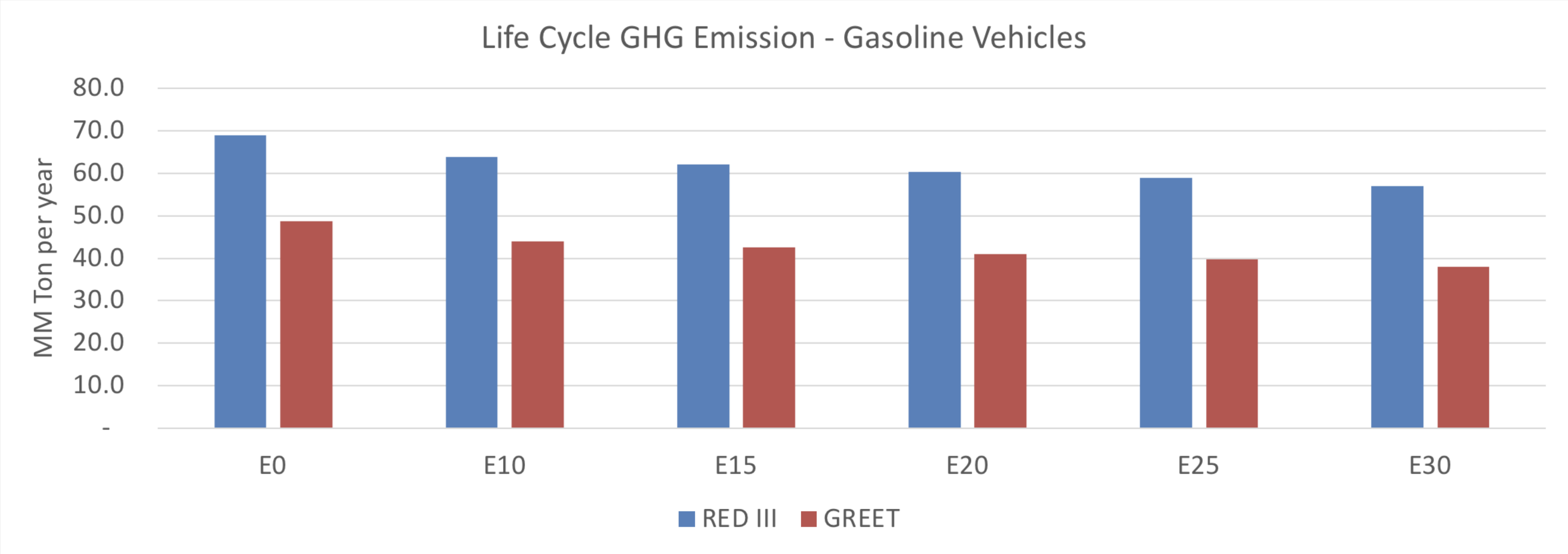
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.





Colombia – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	217.6
2035 Target (MMT/year)	125.5
Delta	92.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.6%	12.5%	15.8%	18.4%	21.9%
RED II/III	0.0%	7.5%	9.9%	12.5%	14.7%	17.4%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	5.1%	6.6%	8.3%	9.7%	11.6%
RED II/III	0.0%	5.6%	7.4%	9.4%	11.0%	13.0%

Source: Greet, RED II/III, climateactiontracker.org, F90